



Influence of technical maintenance measures on ecological status of agricultural lowland rivers – Systematic review and implications for river management

Anna Bącznyk^a, Maciej Wagner^b, Tomasz Okruszko^a, Mateusz Grygoruk^{a,*}

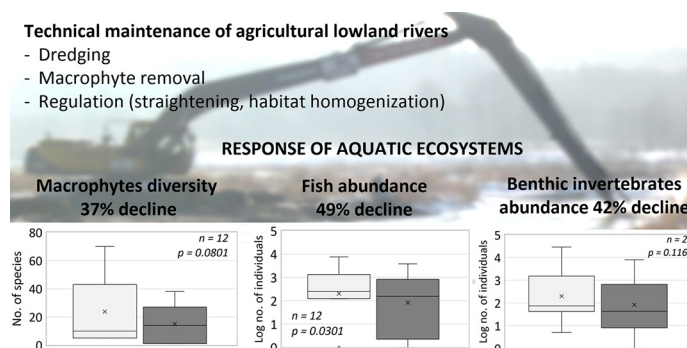
^a Faculty of Civil and Environmental Engineering, Warsaw University of Life Sciences-SGGW, ul. Nowoursynowska 159, 02-776 Warsaw, Poland

^b Institute of Meteorology and Water Management, ul. Podleśna 3, 01-673 Warsaw, Poland

HIGHLIGHTS

- Technical measures of river maintenance induce decline of rivers' ecological status.
- 96% of analyzed studies revealed negative responses of biota to the maintenance of agricultural rivers.
- Maintenance of agricultural rivers induces decline of abundance and species richness of biota.
- Maintenance of agricultural rivers should be revisited in science and better managed in practice.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 15 June 2017

Received in revised form 20 January 2018

Accepted 23 January 2018

Available online 3 February 2018

Keywords:

River
Management
Agriculture
Water Framework Directive
Ecosystem

ABSTRACT

Intensification of agriculture and ongoing urban sprawl exacerbate pressures on rivers. Small rivers in agricultural landscapes are especially exposed to excessive technical actions implemented in order to allow for harvesting river water for irrigation, draining agricultural water and receiving sewage. Regular dredging and macrophyte removal strongly interfere with the global need for preserving river biodiversity that allows agricultural lowland rivers to remain refuges for a variety of species, and—accordingly—to keep water bodies resilient for the benefit of society. In order to provide a comprehensive look at the influence of agricultural lowland river management on the ecological status of these water bodies, we conducted a literature review and a meta-analysis. For the structured literature review we selected 203 papers reflecting on the response of aquatic ecosystems to dredging and macrophyte management actions. The database of scientific contributions developed for our study consists of papers written by the authors from 33 countries (first authorship) addressing dredging, macrophyte removal, status of fish and macroinvertebrates as well as the general ecological status of lowland agricultural rivers. We revealed that 96% of the analyzed papers indicated unilateral, negative responses of aquatic ecosystems, particularly macroinvertebrates, ichthyofauna and macrophyte composition, to maintenance measures. We revealed that studies conducted in the European Union on the ecological status of rivers appeared to significantly increase in quantity after the implementation of the Water Framework Directive. Finally, we concluded that day-to-day management of lowland agricultural rivers requires revision in terms of compliance with environmental conservation requirements and the recurrent implementation of technical measures for river maintenance.

© 2018 Elsevier B.V. All rights reserved.

* Corresponding author.

E-mail address: m.grygoruk@levis.sggw.pl (M. Grygoruk).

1. Introduction

Contemporary water policy in Europe is shaped by the Water Framework Directive of the European Union (WFD; European Commission, 2000). The WFD emphasizes the necessity of sustainable river management with regard to hydromorphology and biodiversity. The WFD also includes regulations on the development and implementation of ecological assessment and classification systems, using biological elements and supportive hydromorphological and physicochemical features to determine a river's ecological status (Leitão et al., 2014; Murat-Błażejewska and Sojka, 2011), providing a framework for improvement of the environmental quality of stream ecosystems, and offering ways for categorizing or measuring ecological status (Baatrup-Pedersen et al., 2008). Balancing ecological quality with agricultural use of catchments and water bodies has become one of the emerging issues during implementation of the WFD. For centuries, humans have transformed landscapes in order to produce food and forage. The extent of landscape modifications differed depending on geographical and climatic settings. A vast majority of watercourses have also become strongly modified (Dynesius and Nilsson, 1994). Water bodies located in agricultural landscapes remain local biodiversity hotspots (Chester and Robson, 2013; Clarke, 2015; Herzon and Helenius, 2008; Watson and Ormerod, 2004). However, the function and ecological status of most rivers in managed landscapes are determined by the frequency of the implementation of river maintenance measures (RMM) such as dredging (mud removal), channel straightening, macrophyte removal or hydromorphological modifications. These RMM were implemented on a regular basis across the lowland landscapes and reported as drivers of biodiversity loss (Adynkiewicz-Piragas and Drabiński, 2001; Armitage and Pardo, 1995; Hachoł and Krzemińska, 2008; Rader and Ward, 1988).

Traditional agricultural water management focuses on reducing the natural variability of river flows to achieve stable water supply and mitigate extreme hydrological events such as floods and droughts. Economic and social motivation for RMM is driven by the need of more intensive landscape productivity; however, a question arises if there are any limits for this activity, taking into account the need to preserve abundance and diversity of riverine communities. A majority of small rivers used as drains/irrigation canals in agricultural landscapes (hereafter referred to as agricultural rivers, ARs) have been modified in terms of their flow regime, in-stream habitat diversity and hydromorphology (Harrison et al., 2004; Johnson, 1988). Similarly, nearly 80% of all water discharge of the largest river systems in both North America and Europe is affected by agriculture and industry-related stressors (Alexandre et al., 2013; Dynesius and Nilsson, 1994). As observed nearly 30 years ago, reducing the diversity and patterning of aquatic habitats in agricultural rivers results in the degradation of river ecosystems (Hellawell, 1988) and endangers the biodiversity of approximately 65% of the world's river habitats, putting thousands of aquatic wildlife species at risk of extinction (Guti and Berczik, 2014). Although legislation systems and knowledge have quickly moved forward in the last decade, and nature-based solutions in RMM have been developed (e.g., Prus et al., 2017) and mandated (*vide* WFD), available scientific observations allow to put a hypothesis that river maintenance has largely remained "traditional" in terms of its scale and continues to influence riverine and riparian ecosystems negatively and across scales (Adynkiewicz-Piragas and Drabiński, 2001; Bierežnoj-Bazille and Grygoruk, 2013; Hachoł and Krzemińska, 2008).

The negative influence of technical RMM on particular elements of riverine and riparian ecosystems of ARs has been repeatedly reported in the worldwide scientific literature since the 1980s (e.g., Murphy et al., 1987; Rader and Ward, 1988; Swales, 1982a, 1982b). However, river-management authorities responsible for implementation of RMM and—at the same time—implementation of the WFD seem not to deal with these actions as with critical stressors for these ecosystems (Smokorowski and Pratt, 2007).

The main goals of the paper are:

- to summarize the most up-to-date basic research on the influence of river maintenance measures on particular elements of riverine ecosystems,
- to identify research gaps and issues related to AR maintenance in terms of compliance with the WFD and claims about sustainability in water-resources management in lowland river basins.

The research goals were formulated in order to provide an opportunity for future development of best management practices which would reduce the threats to the biodiversity of ARs and would ensure their agricultural and ecological functions. Even though the study is driven by a Central European perspective on water management and the status of ARs in the landscape and mesoscale lowland geoeosystems, we believe it provides insights into the international context.

2. Meta-analysis and publication search criteria

This study is based on results of literature meta-analysis. The first step of our approach was oriented at selecting the most comprehensive set of scientific contributions possible. To ensure the highest possible quality of the input data and the conclusions discussed in our analysis, only international peer-reviewed papers listed in the Web of Science Core Collection, SCOPUS and EBSCO databases were analyzed. The research reviewed was published in English. We are aware that results of the query did not cover all research results published in the given field and in national languages, but in order to perform the analysis based on homogeneous criteria, we decided to limit the database to items that have been peer-reviewed and published in recognized books and journals. These databases were subjected to two structured queries. The first query included combinations of keywords: TI = fish* OR macrophyte* OR macroinvertebrate* AND dredg* OR removal OR regulat* AND TS = water management. The second query included combinations of keywords: TI = (fish OR macrophyt* OR macroinvertebrate*) AND TS = water management AND TI = lowland river. This query allowed to select relevant studies addressing the issues of ecological status of managed lowland agricultural rivers. The asterisk inserted at the end of the search terms enabled all variations of the key word to be included in the search. As the research was aimed at revealing a general influence of RMM on fish, macroinvertebrates and macrophytes, the set of papers retrieved by these queries was filtered with the inclusion and exclusion criteria indicated in Table 1. Studies retrieved with the use of an automated query that clearly did not match the criteria referring to lowland rivers (e.g. studies done in upland and mountainous streams) were removed from the database. The resulting database of papers and their features (year of publication, author's origin, number of citations) was used as a set of input data for the meta-analysis (see References and additional literature presented in Appendix 1).

In order to address the research questions, we conducted quantitative analysis supported by standard statistical testing of differences with a *t*-test, with special focus on before-after and control-impact (BACI) studies dealing with the response of aquatic ecosystems to RMM such as dredging, macrophyte removal and broadly considered channel regulation. We searched for publications providing information on four basic RMM, namely channel regulation, mechanical dredging, macrophyte removal (weed cutting) and removal of debris jams, and describing impacts of these RMM on aquatic ecosystems (Table 2).

3. Results and discussion

3.1. Trends in publications

For the review and meta-analysis, we selected 203 papers which met inclusion criteria given in Table 1. Among the selected papers

Table 1
Criteria for inclusion and exclusion of reviewed publications.

Criteria	Include	Exclude
Ecosystem	Lowland river	Estuaries, lakes, coastal waters, groundwater, reservoirs, sea, shelf, dryland
Location	Global (all countries over the world)	None
Project topics	Fish, macroinvertebrate, macrophytes, benthos, dredging, removal, regulation, water management	Climate change, pollution (CO ₂ , carbon, methane, phosphorus, nitrogen), ecosystem services, chemistry

were 65 contributions about ichthyofauna, 75 about macrophytes (including macrophyte removal), 59 about macroinvertebrates and 4 about mammals and birds.

A vast majority of studies were conducted by national teams. The authors of the selected publications represented 33 countries (Fig. 1A). Only 6% of the analyzed papers were published by international teams. Nearly 19% of the selected papers were published by authors from the UK, which—together with the number of papers published by US authors (13%)—constitutes nearly a quarter of the worldwide literature on the topic. The papers retrieved were published between 1976 and 2017. The number of papers continuously increased from 1976 to 2015 (Fig. 1D). Surprisingly, the number of papers representing hydromorphological changes resulting from RMM decreased over the analyzed period (Fig. 1B). Interestingly, the number of papers dealing with multiple aspects (e.g. both macrophytes and fish) increased, which may represent an increasing scientific interest in the integrated analysis of aquatic ecosystems' responses to technical RMM (Fig. 1B). Over the years, most of the papers were published by authors from the European Union member states (EU; Fig. 1C; "EU" refers to the set of member states as of 2017). As expected, after the resolution of the WFD, European interest in the research of the influence of RMM on the ecological status of aquatic ecosystems significantly increased (Fig. 1E). Such an observation reflects a strong influence of international environmental policy on formulating research questions and hypotheses.

This may also represent the importance of research on river ecosystems in the EU: after the resolution of the WFD, research of the ecological status of rivers became a high priority. Mean time span of sampling in the analyzed pre- and post-monitoring studies on the response of macroinvertebrates and fish to RMM reached only about 1 year on average (Fig. 2). The most frequent cases presented investigations of riverine ecosystems between 1 and 7 months before and not more than 14 months after the implementation of RMM, normally covering a span of a couple of months between pre- and post-monitoring. This is likely a result of the duration of scientific projects, lasting normally from 1

to 3 years, which do not allow scientists to continue their research when financing related to given research projects is over. Only four articles reported on long-term studies (lasting over 10 years; some 2%; Brooks et al., 1989; De Jalon et al., 1994; Fjellheim and Raddum, 1996; Holt et al., 2015). In some studies (such as Gorzel and Kornijów, 2007; McCabe et al., 1988; Zawal et al., 2016), multiple sampling campaigns were performed in order to reveal the dynamics of re-establishing populations of biota in disturbed systems.

96% of the analyzed studies presented a significantly negative influence of technical RMM such as dredging (mud removal, shaping longitudinal profile of the river bed), macrophyte removal or channel regulation on ecological status of AR. Only 4% of papers reported no significant negative effect on the ecological status of rivers or reported its enhancement. However, those studies reflected positively on increased abundances of non-native species or increasing numbers of individuals of certain species that—despite the extinction of other species—resulted in a general increase in the number of individuals (mainly macroinvertebrates).

Our analyses indicate that the interest in research on AR ecosystems' responses to RMM may depend on the level of growth and development of particular countries. Although we did not find a correlation between the GDP per capita and the number of papers published by the authors from particular countries (according to the first author's affiliation; Fig. 2A), we observed a significant difference between the numbers of papers published by authors from countries with a GDP per capita higher and lower than 40,000 USD (Fig. 3).

3.2. Impacts

3.2.1. General overview

Agricultural rivers are exposed to continuous stress resulting from RMM (Coeck et al., 1993; Guareschi et al., 2014; Hering et al., 2006; Kruk, 2007). The set of analyzed RMM—namely cutting of weeds, removal of littoral vegetation, removal of woody debris from the bed and banks of rivers, removal of natural and artificial barriers, demolition or modification of beaver dams, and particularly the dredging of sediment and modification of the river channel—affects macrophytes, ichthyofauna and benthic macroinvertebrates (Dąbkowski et al., 2016; Petts, 1988; Petts et al., 1993; Swales, 1982a). As observed, these activities likely have long-lasting effects and result in the alteration of river ecosystems mainly by inducing a discontinuity in the river continuum, loss of spatial habitat heterogeneity and diversity, an increase in water turbidity, a reduction in the abundance of microhabitats, elimination of spawning habitats and the destruction of mussels and other organisms that function in a commensalism-based relation with the others (Crook et al., 2015; Rader et al., 2008). Moreover, uniform habitats in traditionally maintained rivers provide suitable conditions for eurytopic

Table 2
Lowland river maintenance measures addressed in the reviewed papers.

Type of river maintenance measure	Methodology	Impacts	Description references
River regulation	Widening and deepening of the river channel	Changed flow conditions; homogenization of habitats	Gehrke and Harris, 2001; Kruk, 2007; Munn and Brusven, 1991; Rader and Ward, 1988; Swales, 1988; Taylor et al., 2008
	Straightening of the river channel	Changed flow conditions; cut-off of oxbows, homogenization of habitats, increased vulnerability to extreme flows	
Dredging	Mechanical removal of 0.2–0.3 m of the bottom sediments (excavation); profiling river banks.	Removal of individuals of bottom-dwelling organisms, changed flow conditions, increased water turbidity, homogenization of mesohabitats	Aldridge, 2000; Dąbkowski et al., 2016; Hachó and Bondar-Nowakowska, 2012; Ohimain et al., 2005; Rechulicz et al., 2004; Zawal et al., 2016
Weed cutting	Different scenarios of in-channel macrophyte mowing and removal Mowing and removal of bank macrophytes	Affected thermal variability of the river, disruption of fish and fry habitats Affected flow conditions, affected thermal variability of the river, disruption of fish and fry habitats	
Removal of debris jams	Destruction of beaver dams and dead organic matter jams; removal of fallen trees	Homogenization of mesohabitats	Butler and Malanson, 2005; Czech, 2005

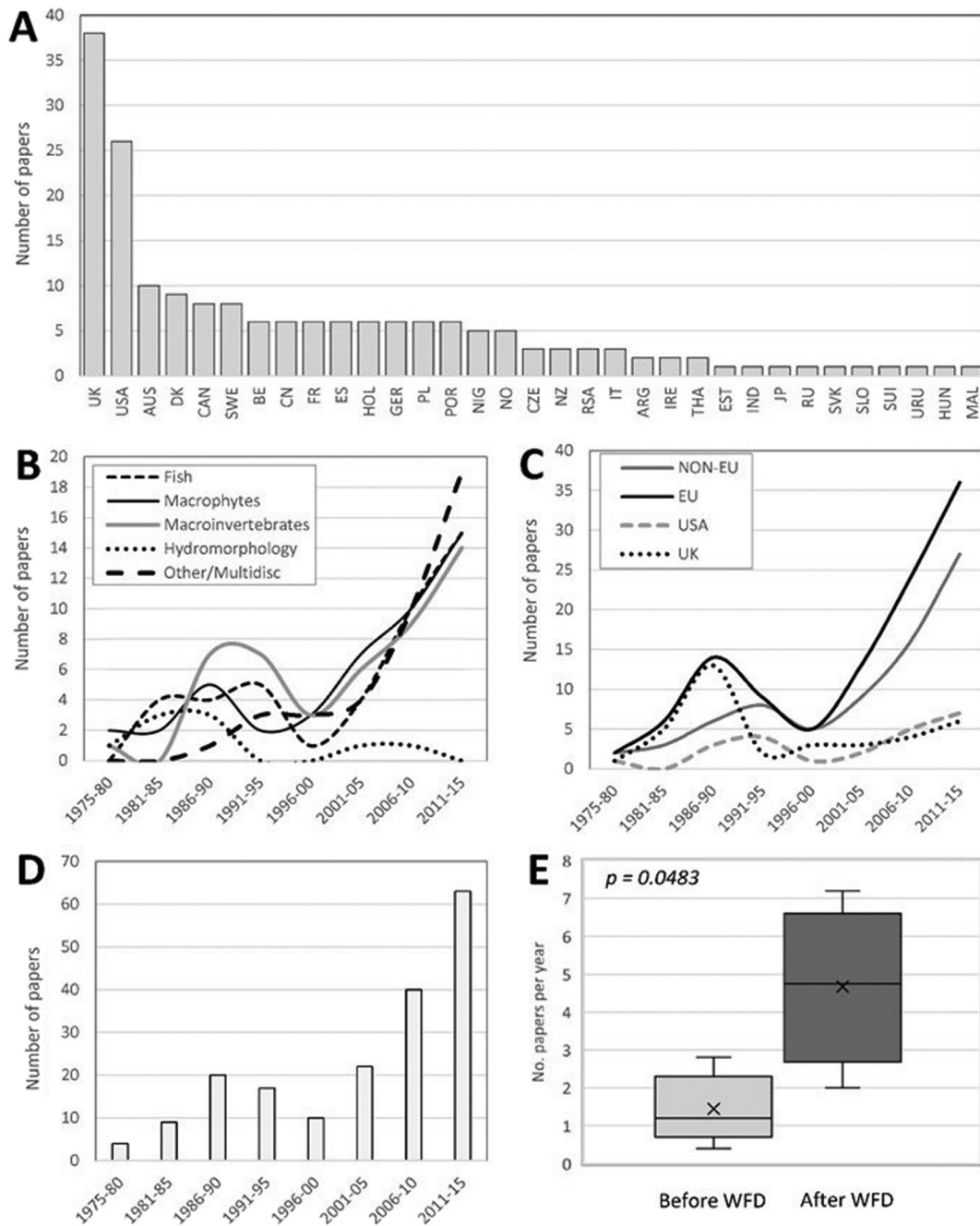


Fig. 1. A – Number of papers published in 1975–2017 according to the country of origin of the first author; B – number of papers dealing with particular elements of riverine ecosystems (1975–2015); C – number of papers published by the authors from EU and non-EU countries (EU referred to as its status in 2017) and extraction of number of papers published by UK and USA authors (1975–2015); D – total number of papers published in 1975–2015; E – comparison of the number of papers published per year in EU countries (EU referred to as its status in 2017) before ($n = 36$) and after ($n = 92$) implementation of WFD (p value of the two-tailed heteroscedastic t -test).

species and force stenotopic species to migrate or go extinct. Habitat homogenization-induced shifts in community composition can, therefore, occur as direct (immediate) and indirect processes caused by interactions among species with different resilience capacities. Along with the nonlinearity of recovery trajectories, these issues were discussed by Cortes et al. (2002) and Kaenel and Uehlinger (1999).

As indicated above, 96% of examples of technical RMM were followed by negative responses in terms of numbers of species/individuals or the area of mesohabitats. It is also known that biocenotic responses to RMM can also occur some years after the application of particular RMM (Petts et al., 1993). Multiple studies revealing the issue of negative environmental consequences of RMM claimed for the need of mitigation of these impacts by introducing modifications to

traditional measures mainly by limiting spatial extents of single actions (e.g., reducing lengths of river stretches to be dredged at a time) and repeated monitoring of riverine ecosystems (Manap and Voulvoulis, 2016; Old et al., 2014; Walker, 1985).

3.2.2. Macrophytes

Among the 12 studies dealing with the BACI analysis of macrophytes' response to river-maintenance activities that are analyzed in our review, we found a 37% decrease in species diversity (from an average of 24 species before the implementation of RMM to an average of 15 after RMM were implemented; Fig. 4A). The decline, although clearly distinguishable, was not statistically significant (t -test, $p = 0.08$).

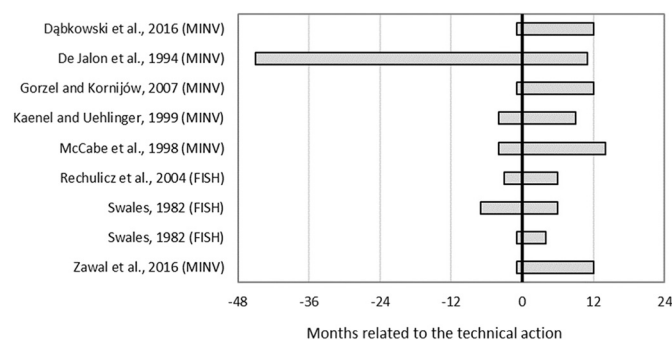


Fig. 2. Duration between pre- and post-inventories in selected BACI studies. MINV – study on macroinvertebrates; FISH – study on ichthyofauna. 0 at the X-axis of the graph stands for the time of implementation of RMM.

Aquatic macrophytes play a critical role in many freshwater ecosystems interacting with flow velocities and affecting sediment accumulation (patchy structure of macrophytes) and light transfer, and thus they maintain fish productivity by stimulating invertebrate production and providing habitat diversity (Baattrup-Pedersen and Riis, 1999; Dar et al., 2014; Hachoł and Bondar-Nowakowska, 2012; Schoelynck et al., 2017). Macrophytes increase structural diversity of environmental conditions (water temperature, light intensity, flow velocity), enhancing maintenance of food webs and services, influencing the production of organic matter, affecting the balance of O_2 , CO_2 , pH and salinity, promoting self-regulation of watercourses and reducing the intensity of bed and/or bank erosion (Bondar-Nowakowska and Hachoł, 2013; Brix and Schierup, 1989; Dar et al., 2014; Dawson and Kern-Hansen, 1978; Franklin et al., 2008; Janauer et al., 2008; McGurk, 1989). Both aquatic macrophytes and riparian vegetation are treated as bioindicators of medium- to long-term ecological changes in water quality (Bondar-Nowakowska et al., 2013; Dallas, 2013; Dar et al., 2014). The importance of macrophytes in biological assessments of rivers is formally recognized in the WFD, and this group of organisms is an essential element in the monitoring of ecological state of rivers (Bus, 2010; Hrivnák et al., 2012; Sierka and Tomczak, 2013; Szoszkiewicz et al., 2010).

Among the methods of macrophyte management, there are nutrient, biological, chemical and mechanical interventions (Johnstone, 1986). Macrophyte mowing and removal has long been known as, in most of the cases, the cheapest (Grygoruk et al., 2014), efficient and thus most frequently applied method to mitigate the growth of aquatic plants (Morgan and Maitland, 1997). Macrophyte cutting, mowing, raking,

digging or pulling decreases hydraulic roughness, allowing to mitigate the flood hazard (Trepel et al., 2003). These activities must be repeated several times to eliminate new growth, and they are often used in rivers draining agricultural lands (Barko et al., 2014). It is known that the response of aquatic vegetation to RMM may be nonlinear, resulting in changes to hydrological (flood velocities alteration), geomorphological (enhancement of sedimentation processes) and biological conditions (serving as habitat for fish) of particular river (Baattrup-Pedersen et al., 2005). It is scientifically substantiated that weed-cutting, dredging and river channel profiling alter the environments of streams and contribute to a significant decrease in macrophyte biomass (Baattrup-Pedersen and Riis, 1999; Baattrup-Pedersen et al., 2003; De Jalon et al., 1994; Lubke et al., 1984; Pedersen et al., 2011).

RMM were reported to influence macrophyte communities and induce the natural succession of plants, resulting in long-term effects on communities' structure (Baattrup-Pedersen et al., 2002; Hachoł et al., 2008; Hachoł and Bondar-Nowakowska, 2012; Kenny and Rees, 1996; Lubke et al., 1984). Annual weed-cutting resulted in directional changes in macrophyte composition—for example, the predominance of eurytopic species and the elimination of stenotopic ones, eventually resulting in homogenization of macrophyte communities (Baattrup-Pedersen and Riis, 2004; de Nie, 1984). Species decline and low macrophyte patch complexity were explained by the intolerance of several species to frequent disturbances and have resulted in a shift toward species-poor patches in frequently disturbed habitats (Baattrup-Pedersen and Riis, 2004; Kaenel and Uehlinger, 1999). The composition and biomass of vegetation after RMM implementation depend on the location, habitat conditions and migration of species from upstream reaches of rivers and from neighboring agricultural areas (Maćkowiak et al., 2014). However, the time of the recovery of aquatic macrophytes after weed-cutting is counted in weeks (Baattrup-Pedersen et al., 2003; Dawson, 1989; De Doncker et al., 2007; Murphy et al., 1987). Some authors state that RMM fragment the environment and interrupt connections between elements of the ecosystem, which increases the diversity of habitats (Hachoł and Bondar-Nowakowska, 2010; Maćkowiak et al., 2014). In the search for a sustainable integral method of controlling aquatic vegetation, there is a need for a better understanding of in-stream ecological relationships (Baattrup-Pedersen and Riis, 1999; Vereecken et al., 2006). However, many papers claimed that to provide optimal conditions for macrophyte communities, technical maintenance practices should cease or—if cessation is not possible—be minimized (Baattrup-Pedersen et al., 2002; Pepper and Rickard, 2009; Iversen et al., 1993).

3.2.3. Macroinvertebrates

Among the 28 studies dealing with the BACI analysis of benthic macroinvertebrates' response to the river maintenance analyzed in our review, we found a 42% decrease in abundance (statistically insignificant; t -test, $p = 0.12$; Fig. 4C). In the case of macroinvertebrates, we found an increase in species diversity (from an average of 29 species before the implementation of river maintenance measures to an average of 33 after the RMM were implemented; Fig. 4D), which was, however, not significant statistically (t -test, $p = 0.66$).

An influence of technical RMM on different aspects of macroinvertebrates' community composition, function, diversity and recolonization abilities were first studied in the 1970s–1980s (Armitage et al., 1987; Boon, 1988; Frugot, 1991). Interest in this important element of riverine ecosystems has been continuously increasing until now (Grygoruk et al., 2015; Guareschi et al., 2014; Ożgo et al., 2016; Zawal et al., 2016). Macroinvertebrate communities constitute one of the most important parts of AR biocenosis due to a number of functions they play in food chains and taxonomic differentiation (Munn and Brusven, 1991). These communities are useful for assessing ecological status and water quality because of their confinement to the river bed and limited movement abilities (Dallas, 2013; Elliott and Drake, 1981; Holt et al., 2015; Pan et al., 2015). Changes in the quality and quantity of benthic

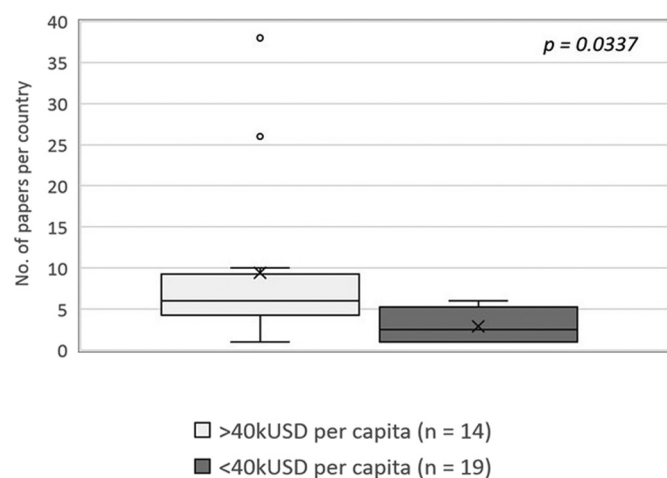


Fig. 3. Comparison of number of papers published in the years 1975–2017 according to the first authors' country's GDP per capita (compared countries of GDP higher and lower than 40kUSD/capita). Data on GDP per capita – World Bank (2015).

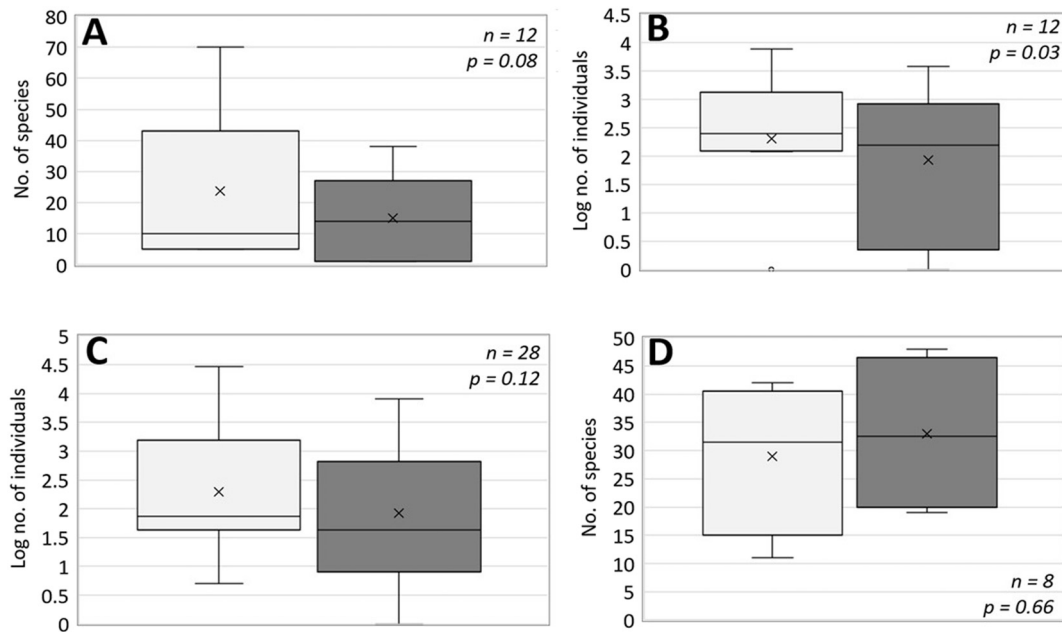


Fig. 4. Comparison of selected BACI studies results (light grey stands for conditions before and dark grey for conditions after the implementation of RMM: A – number of macrophyte species; B – number of fish individuals (values log-transformed); C – number of individuals of macroinvertebrates (values log-transformed) and D – number of macroinvertebrate species.

macroinvertebrates provide clues to the understanding of the environmental “health” of the affected river stretch (Armitage and Pardo, 1995). The appearance of macroinvertebrates depends on the structure of the river bed, bed sediments, flow velocity distribution, water quality, presence of waste, temperature, microhabitat and detrital input (Rader and Ward, 1988). All of these factors, which determine the abundance and vital conditions of macroinvertebrates, are somewhat affected by technical RMM. However, the responses of macroinvertebrate communities to channel regulation are very complex and can vary according to the type of habitat and the scale of stress (Fjellheim et al., 1993). Nevertheless, it may be concluded that RMM is a considerable disturbance and that its effects on biological recovery are prolonged (Kenny and Rees, 1996).

Studies on the effects of AR regulation and dredging indicate that these maintenance actions firstly result in the decline of macroinvertebrate abundance and secondly, in decline of diversity as only the resilient and eurytopic species remain present. These engineering interventions also result in prolonged high water turbidity, the reduction of sediment nutrients, physical disturbances (e.g. alterations in temperature) and physiological and toxic stress (Fjellheim et al., 1993; Guareschi et al., 2014; Haxton and Findlay, 2008; Munn and Brusven, 1991; Ohimain et al., 2005; Ożgo et al., 2016; Principe and del Corigliano, 2006; Rader and Ward, 1988; Rader et al., 2008; Szlauer-Łukaszewska, 2015; Voelz and Ward, 1990). The severity of the listed alterations to the composition and abundance of macroinvertebrate communities depends on the type of RMM and hydromorphological changes they make in the river—for instance, substrate characteristics and instream hydraulics (Benitez-Mora and Camargo, 2014; Fjellheim et al., 1993; Navarro-Llacer et al., 2010; Petts et al., 1993). There are some studies that found no significant effects of RMM on the abundance and structure of benthic invertebrates (Clarke and Tully, 2014). Some of the authors reported the increase in abundance, density and variety of benthic macroinvertebrates in river stretches exposed to the mechanical maintenance works (or downstream from dams; Benitez-Mora and Camargo, 2014; Dukowska et al., 2007; Nilsson and Jansson, 1995; Poff and Zimmerman, 2010; Rader and Ward, 1988; Zawal et al., 2016). Even though the formerly established, natural habitat conditions are somewhat changed by dredging or macrophyte removal, RMM are responsible for creating habitats and niches for new, predominantly

eurytopic species that are frequently more abundant than the species that were formerly present in abundance (Feld et al., 2014; Płaska et al., 2016).

3.2.4. Ichthyofauna

Among the 12 BACI studies aimed at quantifying the response of ichthyofauna to RMM analyzed in our review, we found a statistically significant (t -test, $p = 0.03$) decline in fish abundance that occurred as a result of RMM (Fig. 4B). The average decrease in fish abundance was as high as 49%.

Fish are important components of the food web, encompassing interactions with biotope, phytoplankton, epiphytes, grazers and aquatic macrophytes (de Nie, 1984). As the most economically important element of riverine ecosystems, fish are frequently referred to in environmental policy as indicators of biodiversity and ecological quality (N'Zi et al., 2015).

Yet in the 1980s (Swales, 1982a, 1982b, 1988; Mann, 1988), researchers claimed that a wide range of negative consequences to fish diversity and abundance in managed ARs were likely attributable to maintenance- and channel regulation-driven river habitat modification (Moyle, 1995; Paukert et al., 2008; Taylor et al., 2008). Macrophyte removal in particular and the loss of abundance and diversity of aquatic plant species in general negatively influences the ichthyofauna of AR river systems (de Nie, 1984). Water-level fluctuations that occur in regulated rivers negatively influence the growth of lowland fish species (Gardner et al., 2015): (1) channelized sections of rivers suffer from the lack of refuges for fish during low- and high-flow periods, which results from uniform shape of the river channel, and (2) artificially controlled water levels and discharges (and hence, flow velocities) influence the shift of seasonal patterns of in-stream and riparian habitats (e.g. Marcinkowski and Grygoruk, 2017). Modification and regulation of flow regime in floodplain-related ARs affect fry recruitment by reducing the ability of fish to access their spawning grounds in a particular floodplain (Bain et al., 1988; Coeck et al., 1993; Gehrke et al., 1995; Gehrke and Harris, 2001; Humphries and Lake, 2000; Humphries et al., 2002; Jurajda, 1995; Macnaughton et al., 2014; Pool and Olden, 2015; Rabeni and Jacobson, 1993; Swales, 1988). By critically changing the structure of the bed sediment in a few kilometers-long river stretches, dredging negatively affects spawning conditions for rheophile fish

species (Freedman et al., 2013; Smokorowski and Pratt, 2007; Wiśniewolski and Gieriej, 2011), causing a simultaneous reduction in abundance, biomass and community diversity, changing the fish assemblage's structure, disordering fish life cycle, preventing spawning, and reducing/eliminating recruitment (Punchard et al., 2008; Taylor et al., 2008). Mechanical dredging and weed-cutting are normally reported as causing shifts in macroinvertebrate species, leading to fish starvation because of the reduction in food resources (Aldridge, 2000).

Hydrotechnical structures in AR (weirs, culverts, spillways) are also reported to distort the status of ichthyofauna by negatively influencing longitudinal connectivity of the river continuum, thereby affecting fish migration (especially anadromous species; Gardner et al., 2015; Obande et al., 2014) and often resulting in the total closure of the migratory corridor (Crook et al., 2015; Wiśniewolski and Gieriej, 2011).

Some papers discussed the lack of differences in conditions of fish populations (e.g. growth), which could be attributed to AR management and maintenance (Brooks et al., 1989; Crisp et al., 1983). Similar to the case of benthic macroinvertebrates, some studies concluded that channel regulation and maintenance induce the development of habitat conditions suitable to species that did not previously occur in a particular water body. It has been observed that hydromorphological pressures on rivers, together with water pollution, resulted in the conversion of fish species structure, from that typical of rivers dominated by salmonids to one dominated by chub *Squalius cephalus*, roach *Rutilus rutilus* and gudgeon *Gobio gobio* (Teixeira-de Mello et al., 2015).

4. Implications for future research and river management

In the light of the presented systematic review, one can conclude that technical maintenance of AR such as dredging, channel regulation (straightening and widening of riverbed) and macrophyte removal result in an unilateral, negative response of fish, macroinvertebrates and macrophytes. Hence, AR maintenance—in general—contradicts the requirements of the European Union WFD and associated regulations such as the EU Habitat Directive. In the face of the negative consequences of RMM on riverine ecosystems, their implementation should be reconsidered, accounting for the potential gain in agricultural production and implemented responsibly. Expecting the abundances and diversity of fish, macroinvertebrates and macrophytes to decline as a result of RMM, it is compulsory to analyze trajectories of the river's ecological response (Fig. 5). Most of the available studies refer to the response of biota to a single “technical measure” stress (e.g. Dąbkowski et al., 2016; Kaenel and Uehlinger, 1999; Rechulicz et al., 2004; Swales, 1982a; Zawal et al., 2016). Hence, the analysis of the responses of particular elements of riverine ecosystems remains limited to a certain, short period of time (Fig. 6; grey field in the figure), not addressing the issue of numerous alternative states of the ecosystem (marked in Fig. 5). Among the analyzed research, only the study of De Jalon et al. (1994) provided the reference “before” sampling approximately 45 months prior to RMM implementation. Most likely the short time span of most of the reviewed BACI studies (Fig. 2.; from less than 3 months on average for ‘before’ inventories, to 10 months on average for the “after” comparison) is related to limited duration of research projects. Field sampling of biota in AR subjected to regular RMM also depends on strict temporal regime of the river maintenance that does not allow finding appropriate reference (Zawal et al., 2016). Although such analyses are valuable, being the only possible evidence-based BACI study designs, they are likely not to represent the response of AR in a comprehensive manner. The analyzed responses consider the status from ‘before’ the stressor occurrence as a reference state, which in the case of rivers subjected to a regular former dredging or macrophyte removal is not justified. As proven, the self-recovery of stressed populations of macroinvertebrates (Gorzel and Kornijów, 2007; Monahan and Caffrey, 1996), macrophytes (Rader and Ward, 1988) or fish (Rechulicz et al., 2004) does not follow linear trajectories. Moreover, response trajectories strongly depend on the resilience of

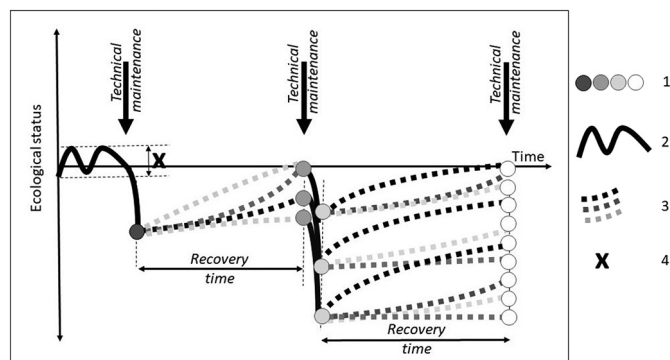


Fig. 5. Hypothetical nonlinear responses of ecological status of maintained rivers. 1 – particular alternative states of ecosystems of the maintained river at a time; 2 – variability of ecological status and known decline of ecological status caused by RMM; 3 – hypothetical response trajectories; 4 – natural variability of river ecological status related to its resilience and disturbances;

species and longitudinal connectivity of riverine ecosystems. Hence, most BACI design studies are only representative for single recovery trajectories or single, time- and space-limited combination of alternative states (widely discussed by Scheffer et al. (2009) and Suding and Hobbs (2009)). Therefore, knowledge gaps in assessing response trajectories (Fig. 5. and Fig. 6: β and γ) as a function of time (e.g. analysis of stressors from multiple years) and space (e.g. stressors implemented in different stretches along the river course) result from limited consideration of alternative states of riverine ecosystems. Hence, consideration of a ‘before’ status as a reference state does not account for former responses to historical stressors nor for the natural fluctuation of biocenotic features (referred to as x , Fig. 6). Some sampling-based studies found the recovery time to be ‘short’ (counted in months and years; e.g. Buczyński et al., 2016; Petts et al., 1993). However, the lack of relevant, multi-annually documented alternative states and knowledge of former response trajectories allows considering AR ecosystem recovery an issue to be addressed in the future research.

Another issue that we found inappropriately addressed in the reviewed studies is the capability of self-recovery to reach certain ecological resilient stability, different from the one considered as a reference (Fig. 6., index A). Most likely, however, the full process of self-recovery is impossible to be quantified in managed AR as the recurrence time of RMM implemented in AR is short (once a year; once a couple of years; Fig. 6). We hypothesize that in certain site-specific conditions, one single application of RMM (e.g. short stretch-based dredging of a part of river channel) might result in the improvement of certain indicators of river ecological status. In that case, A (Fig. 6) could reach the level higher than x . Unfortunately, the numbers of available BACI studies did not allow to constitute the hierarchy of the level of influence of RMM on the ecological status of AR. Moreover, intervals between the implementation of RMM are shorter than the recovery time in all of the studies reviewed. Therefore, the hypothesis about maximum level of ecosystem recovery (A – Fig. 6.) higher than the reference, natural variability of ecological status (x – Fig. 6.) requires to be addressed in possible future research.

Dredging and channel regulation (Manap and Voulvoulis, 2016; Ohimain et al., 2005; Wenger et al., 2017) seem to be more influential on fish, macroinvertebrates and macrophytes than some ‘light’ measures such as macrophyte mowing and removal (Bal and Meire, 2009; Old et al., 2014; Pedersen et al., 2011). On the other hand, some papers indicating ambiguous responses of certain riverine ecosystems even to “hard” stressors such as deep dredging (Branco et al., 2012; Crisp et al., 1983; Zawal et al., 2016) do not allow these observations to be unilaterally extrapolated to other river systems.

Our conclusions about adequate monitoring of RMM influence to AR can be transferred to the monitoring applied in those river restoration projects that require technical measures to be implemented. Technical

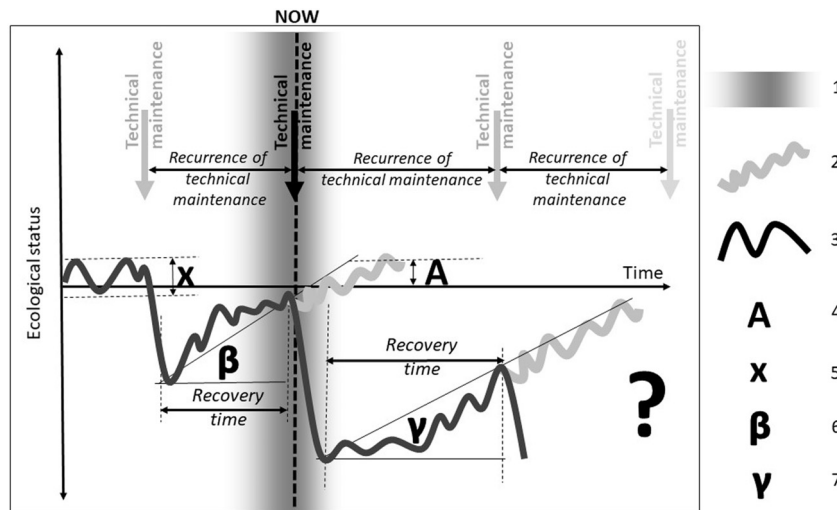


Fig. 6. Dynamics of ecological status of rivers exposed to RMM. 1 – time span of the reviewed studies; 2 – probable trajectory of self-restoration of ecological status if RMM was ceased; 3 – variability of ecological status and known decline of ecological status caused by RMM; 4 – hypothetical maximum level of ecosystem recovery if the RMM was not implemented; 5 – natural variability of river ecological status related to its resilience and disturbances; 6 – angle of n th ecosystem recovery trajectory; 7 – angle of $n + 1$ st ecosystem recovery trajectory. The figure presents selected, particular and hypothetical trajectory of the integrated “ecological status”.

actions aimed at reconnection of formerly disconnected floodplain lakes to the main river channel and opening abandoned side arms, although aimed at the improvement of the environmental quality, are expected – similarly to the reviewed RMM – to result in short-term negative responses of fish, macrophytes and macroinvertebrates. Some of the floodplain lakes could have been artificially disconnected from the main river channel for long time (many channel regulation activities in Europe took place in the nineteenth century) (e.g., [Deoniziak et al., 2017](#)). This situation could have induced river bed incision resulting in modification of the water level-discharge relationship (lowering of water level at the same discharge). In such cases, reconnection of oxbows would require dredging and deepening of floodplain lakes in order to achieve the desirable duration of their hydraulic connectivity with the main river. Reconnection of such floodplain lakes would therefore require excessive technical interventions, which – in the light of our results – are likely to be followed by negative responses of the riverine environment. In this case, restoration projects which are usually well structured and comprehensively monitored, may allow to study recovery trajectories of stressed systems, providing data to interpret A , β and γ (Fig. 5).

Facing the issues addressed in our analysis, we claim that day-to-day management of lowland agricultural rivers requires revision in terms of compliance with the environmental conservation policies and the recurrence of implementation of technical maintenance measures in ARs. Results of our review and concepts on the dynamics of river ecological status (Fig. 5. and 6.) indicate that policies regulating RMM in AR should aim at introducing and promoting ‘lighter’ measures (weed cutting; although its negative influence is known in terms of limiting macroinvertebrate biodiversity and fish recruitment) and limiting ‘hard’ measures by (1) increasing the recurrence intervals between the technical maintenance actions and (2) decreasing lengths of maintained stretches allowing to keep their ecological status at desirably high level by promoting re-colonization and reduction of disturbances to aquatic and riparian ecosystems.

5. Conclusions

Ongoing, intensifying and expanding agriculture has been defined one of the critical threats to the environment in the twenty-first century ([Giovannucci et al., 2012](#)). Consequently, agricultural lowland rivers that become inherent elements of agricultural landscapes will remain under increasing pressures originating from the need of managing

them for agriculture. Despite environmental regulations putting priorities on healthy environment, economic pragmatism is likely to underpin – as it has been observed up to now and documented in our review – new challenges to aquatic ecosystems. Ecosystems of AR considered local biodiversity hotspots will be endangered, if traditional and invasive RMM continue to be implemented on a regular basis with no regard to status and potential of water bodies. 96% of studies reviewed in our paper reported unilateral, negative responses of aquatic ecosystems to RMM, indicating qualitative and quantitative changes in the abundance and composition of the flora and fauna. The analyzed studies reported overall decline in the abundance of macrophytes and benthic macroinvertebrates, and statistically significant decline of fish abundance in rivers subjected to RMM. Such results indicate that studies on integrated responses of macrophytes, macroinvertebrates and fish to invasive RMM, although revealed to be increasing in numbers in recent years, are still required. Additionally, long-term (5-, 10- and 20-year) BACI studies on the response of AR ecosystems to technical maintenance measures are lacking. Thus, the resilience of aquatic ecosystems persisting under regular stress from changing flow conditions (macrophyte removal, broadening and straightening of the river course) and sediment structure (dredging), although intuitively obvious, can hardly be quantified on the basis of available research.

Similarly, trajectories of self-restoration of AR ecosystems after dredging, channel regulation or macrophyte removal are not yet known and as such should become an objective of the future research. It is expected that in the case of recurrent implementation of technical river-maintenance actions, particular elements of the riverine and riparian environment will recover at different rates and will most likely go in different directions (i.e., decline of abundance of some stenotopic species and increase of the abundance of eurytopic species; appearance of new species and extinction of the ones that were present before the recurrent application of RMM). Studies reviewed in our study do not allow either to conclude on proven trajectories of self-restoration and temporal dynamics of this process.

Filling the above-listed gaps will, in the authors' opinion, require designing and implementing long-term monitoring programs. As the dynamics of AR ecosystems' responses to RMM are expected to be dependent on the recurrence of RMM, it is highly unlikely that such sampling programs could be covered in the standard frameworks of research projects (normally not more than 3 years). Authors claim that such long-term response-oriented sampling procedures should be developed in the protocols of national bodies responsible for

implementing the WFD. Long-term data sets could then allow scientists to foresee the effects of RMM on biodiversity and verify the traditional role of ARs as local biodiversity hotspots. Only in this way could river-maintenance measures be in order to ensure the sustainable management of small lowland rivers, thus benefitting regional environments and societies.

Hypothetical trajectories of the recovery of river's ecological status after its deterioration resulting from RMM studied in our review allow to state that AR can persist as biodiversity hotspots only if RMM stress was mitigated by incorporating environmental knowledge in day-to-day river management. However, observing slow progress in incorporating environmental features in RMM for the last 30 years, we do not expect this status to change dramatically in the near future. Thus, dredging, weed cutting and debris jams removal will remain important stressors to lowland rivers. We conclude that if economic drivers to RMM implementation persist over time, the only way to mitigate negative influence of RMM to ecological status of AR in a short-time perspective is to keep the recurrence intervals of RMM longer than self- or human-supported recovery of riverine ecosystems. Along, scientists could help in actively providing new insights into the response of AR to RMM by providing longer monitoring datasets and analyzing interconnected responses of biota to RMM-related disturbances.

Finally, we claim that trade-off analyses are required incorporating assessments of hydraulic efficiency of particular RMM and the levels of their negative influence to AR ecological status. Such an integrated research, so far insufficiently addressed in the reviewed literature, will be required when environmental and economic priorities will gain equal importance in decision making in river management. At the time, when ecosystem services influence global paradigms of environmental management (Seppelt et al., 2011), we expect this moment to come. The sooner, the better for lowland rivers persisting in agricultural landscapes of the managed world.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2018.01.235>.

Acknowledgements

We wish to thank Bartłomiej Wyżga for thoughtful comments, edits and inspiring discussions that helped us to structure the paper and formulate its main conclusions. We also acknowledge three anonymous reviewers for thoughtful revisions and comments that greatly helped us to improve the manuscript.

References

- Adynkiewicz-Piragas, M., Drabiński, A., 2001. Wpływ inwestycji hydrotechnicznych na ekosystem rzeki Smortawy. *Zeszyty Naukowe Akademii Rolniczej we Wrocławiu. Melioracja* 43, 7–29.
- Aldridge, D.C., 2000. The impacts of dredging and weed cutting on a population of freshwater mussels (*Bivalvia*: Unionidae). *Biol. Conserv.* 95, 247–257.
- Alexandre, C.M., Ferreira, T.F., Almeida, P.R., 2013. Fish assemblages in non-regulated and regulated rivers from permanent and temporary Iberian systems. *River Res. Appl.* 29, 1042–1058.
- Armitage, P.D., Pardo, I., 1995. Impact assessment of regulation at the reach level using macroinvertebrate information from mesohabitats. *Regul. Rivers Res. Manag.* 10, 147–158.
- Armitage, P.D., Gunn, R.J.M., Furse, M.T., Wright, J.F., Moss, D., 1987. The use of prediction to assess macroinvertebrate response to river regulation. *Hydrobiologia* 144, 25–32.
- Baatrup-Pedersen, A., Riis, T., 1999. Macrophyte diversity and composition in relation to substratum characteristics in regulated and unregulated Danish streams. *Freshw. Biol.* 42, 375–385.
- Baatrup-Pedersen, A., Riis, T., 2004. Impacts of different weed cutting practices on macrophyte species diversity and composition in a Danish stream. *River Res. Appl.* 20, 103–114.
- Baatrup-Pedersen, A., Larsen, S.E., Riis, T., 2002. Long-term effects of stream management on plant communities in two Danish lowland streams. *Hydrobiologia* 481, 33–45.
- Baatrup-Pedersen, A., Larsen, S.E., Riis, T., 2003. Composition and richness of macrophyte communities in small Danish streams – influence of environmental factors and weed cutting. *Hydrobiologia* 495, 171–179.
- Baatrup-Pedersen, A., Freiberg, N., Larsen, S.E., Riis, T., 2005. The influence of channelisation on riparian plant assemblages. *Freshw. Biol.* 50, 1248–1261.
- Baatrup-Pedersen, A., Springe, G., Riis, T., Larsen, S.E., Sand-Jensen, K., Larsen, L.M.K., 2008. The search for reference conditions for stream vegetation in northern Europe. *Freshw. Biol.* 53, 1890–1901.
- Bain, M.B., Finn, J.T., Booke, H.E., 1988. Streamflow regulation and fish community structure. *Ecology* 69, 382–392.
- Bal, K.D., Meire, P., 2009. The influence of macrophyte cutting on the hydraulic resistance of lowland rivers. *J. Aquat. Plant Manag.* 47, 65–68.
- Barko, J.W., Adams, M.S., Clesceri, N.L., 2014. Environmental factors and their consideration in the management of submersed aquatic vegetation: a review. *J. Aquat. Plant Manag.* 24, 1–10.
- Benitez-Mora, A., Camargo, J.A., 2014. Ecological responses of aquatic macrophytes and benthic macroinvertebrates to dams in the Henares River Basin (Central Spain). *Hydrobiologia* 728, 167–178.
- Biereżnoj-Bazille, U., Grygoruk, M., 2013. Scale matters: efficiency assessment of EU environmental directives implementation in a local-scale management of protected wetlands in Poland. *Scientific Annals of the Danube Delta Institute*. 19, pp. 5–12.
- Bondar-Nowakowska, E., Hachoł, J., 2013. Tendencje zmian w zbiorowiskach roślin wodnych w ciekach o różnym sposobie umocnienia skarp. *Infrastruktura i Ekologia Terenów Wiejskich* 3 (1), 205–215.
- Bondar-Nowakowska, E., Hachoł, J., Lubczyński, A., 2013. Wpływ przekształceń koryta na makrofity wodne na przykładzie rzeki Smortawy. *Infrastruktura i Ekologia Terenów Wiejskich* 3 (4), 229–242.
- Boon, P.J., 1988. The impact of river regulation on invertebrate communities in the U.K. *Regul. Rivers Res. Manag.* 2, 389–409.
- Branco, P., Segurado, P., Santos, J.M., Pinheiro, P., Ferreira, M.T., 2012. Does longitudinal connectivity loss affect the distribution of freshwater fish? *Ecol. Eng.* 48, 70–78.
- Brix, H., Schierup, H.H., 1989. The use of aquatic macrophyte in water-pollution control. *Ambio* 18, 100–107.
- Brooks, R.J., Nielsen, P.S., Saltveit, S.J., 1989. Effect of stream regulation on population parameters of Atlantic salmon [*Salmo salar* L.] in the river Lærdalselva, western Norway. *Regul. Rivers Res. Manag.* 4, 347–354.
- Buczyński, P., Zawal, A., Buczyńska, E., Stępień, E., Dąbkowski, P., Michoński, G., Szlauer-Łukaszewska, A., Pakulnicka, J., Stryjecki, R., Czachorowski, S., 2016. Early recolonization of a dredged lowland river by dragonflies (Insecta: Odonata). *Knowl. Manag. Aquat. Ecosyst.* (417), 1–11.
- Bus, A., 2010. Wstępna ocena stanu ekologicznego małej rzeki nizinnej na podstawie makrofytowej metody oceny rzek. *Infrastruktura i Ekologia Terenów Wiejskich* 9, 221–230.
- Butler, D.R., Malanson, G.P., 2005. The geomorphic influences of beaver dams and failures of beaver dams. *Geomorphology* 71, 48–60.
- Chester, E.T., Robson, B.J., 2013. Anthropogenic refuges for freshwater biodiversity: their ecological characteristics and management. *Biol. Conserv.* 166, 64–75.
- Clarke, S.J., 2015. Conserving freshwater biodiversity: the value, status and management of high quality ditch systems. *J. Nat. Conserv.* 224, 93–100.
- Clarke, S., Tully, O., 2014. BACI monitoring of effects of hydraulic dredging for cockles on intertidal benthic habitats of Dundalk Bay, Ireland. *J. Mar. Biol. Assoc. U. K.* 94, 1451–1464.
- Coeck, J., Vandelanoot, A., Yseboodt, R., Verheyen, R.F., 1993. Use of the abundance/biomass method for comparison of fish communities in regulated and unregulated lowland rivers in Belgium. *Regul. Rivers Res. Manag.* 8, 73–82.
- Cortes, R.M.V., Ferreira, M.T., Oliveira, S.V., Oliveira, D., 2002. Macroinvertebrate community structure in a regulated river segment with different flow conditions. *River Res. Appl.* 18, 367–382.
- Crisp, D.T., Mann, R.H.K., Cubby, P.R., 1983. Effects of regulation of the River Tees upon fish populations below Cow Green reservoir. *J. Appl. Ecol.* 20, 371–386.
- Crook, D.A., Lowe, W.H., Allendorf, F.W., Eros, T., Finn, D.S., Gillanders, B.M., Hadwen, W.L., Harrod, C., Hermoso, V., Jennings, S., Kilada, R.W., Nagelkerken, I., Hansen, M.M., Page, T.J., Riginos, C., Fry, B., Hughes, J.M., 2015. Human effects on ecological connectivity in aquatic ecosystems: integrating scientific approaches to support management and mitigation. *Sci. Total Environ.* 534, 52–64.
- Czech, A., 2005. Analiza dotychczasowych rodzajów i rozmiaru szkód wyrządzonych przez bobra oraz stosowanie metod rozwiązania sytuacji konfliktowych. *Instytut Ochrony Przyrody, Kraków*.
- Dąbkowski, P., Buczyński, P., Zawal, A., Stępień, E., Buczyńska, E., Stryjecki, R., Czachorowski, S., Śmietana, P., Szenejko, M., 2016. The impact of dredging of a small lowland river on water beetle fauna (Coleoptera). *J. Limnol.* 75, 472–487.
- Dallas, H.F., 2013. Ecological status assessment in Mediterranean rivers: complexities and challenges in developing tools for assessing ecological status and defining reference conditions. *Hydrobiologia* 719, 483–507.
- Dar, N.A., Pandit, A.K., Ganai, B.A., 2014. Factors affecting the distribution patterns of aquatic macrophytes. *Limnologic. Rev.* 14, 75–81.
- Dawson, F.H., 1989. Ecology and management of water plants in lowland streams. Report of the Freshwater Biological Association. 57, pp. 43–60.
- Dawson, F.H., Kern-Hansen, U., 1978. Aquatic weed management in natural streams: the effect of shade by marginal vegetation. *Verhandlungen der Internationalen Vereinigung fur Theoretische und Angewandte Limnologie* 20, 1451–1456.
- De Doncker, L., Verhoeven, R., Bal, K., Meire, P., Okruszko, T., Świątek, D., Chormański, J., Okruszko, et al., 2007. Impact of vegetation development on the hydraulic characteristics and flow patterns in lowland rivers. *Wetlands: Monitoring, Modelling and Management*. Taylor and Francis, London.
- De Jalon, G., Sanchez, P., Camargo, J.A., 1994. Downstream effects of a new hydropower impoundment on macrophyte, invertebrate and fish communities. *Regul. Rivers Res. Manag.* 9, 253–261.
- Deonizak, K., Hermaniuk, A., Wereszczak, A., 2017. Effects of wetland restoration on the amphibian community in the Narew River Valley (Northeast Poland). *Salamandra* 53, 50–58.

- Dukowska, M., Szczerkowska, E., Grzybkowska, M., Tsydel, M., Penczak, T., 2007. Effect of flow manipulations on benthic fauna communities in a lowland river: interhabitat comparison. *Pol. J. Ecol.* 5, 101–112.
- Dynesius, M., Nilsson, C., 1994. Fragmentation and flow regulation of river systems in the Northern third of the World. *Science* 266, 753–762.
- Elliott, J.M., Drake, C.M., 1981. A comparative study of four dredges used for sampling benthic macroinvertebrates in rivers. *Freshw. Biol.* 11, 245–261.
- European Commission, 2000. Directive 2000/60/EC of the European Parliament and of the council of 23 October 2000 establishing a framework for community action in the field of water policy. *Off. J. Eur. Commun. L* 327 (43), 1–72.
- Feld, C.K., de Bello, F., Doledet, S., 2014. Biodiversity of traits and species both show weak responses to hydromorphological alteration in lowland river macroinvertebrates. *Freshw. Biol.* 59, 233–248.
- Fjellheim, A., Raddum, G.G., 1996. Weir building in a regulated West Norwegian river: long-term dynamics of invertebrates and fish. *Regul. Rivers Res. Manag.* 12, 501–508.
- Fjellheim, A., Havaradstun, J., Raddum, G.G., Schnel, Ø.A., 1993. Effects of increased discharge on benthic invertebrates in a regulated river. *Regul. Rivers Res. Manag.* 8, 179–187.
- Franklin, P., Dunbar, M., Whitehead, P., 2008. Flow controls on lowland river macrophytes: a review. *Sci. Total Environ.* 400, 369–378.
- Freedman, J.A., Carlisle, R.F., Stauffer Jr., J.R., 2013. Gravel dredging alters diversity and structure of riverine fish assemblages. *Freshw. Biol.* 58, 261–274.
- Fruget, J.F., 1991. The impact of river regulation on the lotic macroinvertebrate communities of the lower Rhône, France. *Regul. Rivers Res. Manag.* 6, 241–255.
- Gardner, C.J., Deeming, D.C., Eady, P.E., 2015. Seasonal water level manipulation for flood risk management influences home-range size of common bream *Abramis brama* L. in a lowland river. *River Res. Appl.* 31, 165–172.
- Gehrke, P.C., Harris, J.H., 2001. Regional-scale effects of flow regulation on lowland riverine fish communities in New South Wales, Australia. *Regul. Rivers Res. Manag.* 17, 369–391.
- Gehrke, P.C., Brown, P., Schiller, C.B., Moffat, D.B., Bruce, A.M., 1995. River regulation and fish communities in the Murray-Darling River system, Australia. *Regul. Rivers Res. Manag.* 11, 363–375.
- Giovannucci, D., Scherr, S., Nierenberg, D., Hebebrand, C., Shapiro, J., Milder, J., Wheeler, K., 2012. Food and Agriculture: the future of sustainability. A strategic input to the Sustainable Development in the 21st Century (SD21) Project. United Nations Department of Economic and Social Affairs, Division for Sustainable Development, New York.
- Goźel, M., Kornijów, R., 2007. The response of zoobenthos to "natural channelization" of a small river. *Ecohydrol. Hydrobiol.* 7, 59–70.
- Grygoruk, M., Fiedorczuk, P., Kasjanik, A., Kostecka, A., Grygoruk, J., 2014. Monitoring prac utrzymaniowych i usuwania skutków powodzi realizowanych przez Wojewódzki Zarząd Melioracji i Urządzeń Wodnych w Białymstoku: ocena zgodności działań WZMiUW z prawem unijnym i krajowym, ich konsekwencji dla mieszkańców i środowiska naturalnego oraz podejmowanie interwencji w razie wykrycia nieprawidłowości. Stowarzyszenie Niezależnych Inicjatyw Nasza Natura, Ignatki-Osiedle. 40 pp. URL: http://naszanatura.com.pl/wp-content/uploads/2015/01/nasza_naturaRaport.pdf (accessed on 12.12.2017).
- Grygoruk, M., Frąk, M., Chmielewski, A., 2015. Agricultural rivers at risk: dredging results in a loss of macroinvertebrates. Preliminary observations from the Narew catchment, Poland. *Water* 7, 4511–4522.
- Guareschi, S., Laini, A., Racchetti, E., Bo, T., Fenoglio, S., Bartoli, M., 2014. How do hydromorphological constraints and regulated flows govern macroinvertebrate communities along an entire lowland river? *Ecohydrology* 7, 366–377.
- Guti, G., Berczik, A., 2014. Criteria of sustainable management of large river systems - ecological aspects and challenges of the 21st century. *Opuscula Zoologica Budapest* 45, 95–99.
- Hachó, J., Bondar-Nowakowska, E., 2010. Effect of regulation and maintenance works on aquatic plant communities in the Dobra, Żalina and Żurawka rivers. *Zesz. Probl. Postep. Nauk. Rol.* 548, 157–165 (in Polish).
- Hachó, J., Bondar-Nowakowska, E., 2012. Tendencies in the development of hydromacrophytes after the completion of regulatory and maintenance works in a river bed. *Ecol. Chem. Eng.* 19, 997–1013.
- Hachó, J., Krzezińska, A., 2008. Wpływ regulacji rzeki Smortawy na przebieg procesów samooczyszczania na przykładzie wskaźników tlenowych. *Infrastruktura i Ekologia Terenów Wiejskich* 9, 207–216.
- Hachó, J., Bondar-Nowakowska, E., Reinhard, A., 2008. Oddziaływanie wybranych elementów fizycznych koryta cieków na zbiorowiska naczyniowych roślin wodnych. *Infrastruktura i Ekologia Terenów Wiejskich* 7, 255–266.
- Harrison, S.S.C., Pretty, J.L., Shepherd, D., Hildrew, A.G., Smith, C., Hey, R.D., 2004. The effect of instream rehabilitation structures on macroinvertebrates in lowland rivers. *J. Appl. Ecol.* 41, 1140–1154.
- Haxton, T.J., Findlay, C.S., 2008. Meta-analysis of the impacts of water management on aquatic communities. *Can. J. Fish. Aquat. Sci.* 65, 437–447.
- Hellawell, J.M., 1988. River regulation and nature conservation. *Regul. Rivers Res. Manag.* 2, 425–443.
- Hering, D., Johnson, R.K., Kramm, S., Schmutz, S., Szoszkiewicz, K., Verdonschot, P.F.M., 2006. Assessment of European streams with diatoms, macrophytes, macroinvertebrates and fish: a comparative metric-based analysis of organism response to stress. *Freshw. Biol.* 51, 1757–1785.
- Herzon, I., Helenius, J., 2008. Agricultural drainage ditches, their biological importance and functioning. *Biol. Conserv.* 141, 1171–1183.
- Holt, C.R., Pfitzer, D., Scalley, C., Caldwell, B.A., Batzer, D.P., 2015. Macroinvertebrate community responses to annual flow variation from river regulation: an 11-year study. *River Res. Appl.* 31, 798–807.
- Hrivnák, R., Otáhel'ova, H., Gomory, D., Valachovic, M., Paľove-Balang, P., 2012. Environmental effects on species richness of macrophytes in Slovak streams. *Cent. Eur. J. Biol.* 7, 1030–1036.
- Humphries, P., Lake, P., 2000. Fish larvae and the management of regulated rivers. *Regul. Rivers Res. Manag.* 16, 421–432.
- Humphries, P., Serafini, L.G., King, A.J., 2002. River regulation and fish larvae: variation through space and time. *Freshw. Biol.* 47, 307–331.
- Iversen, T.M., Kronvang, B., Madsen, B.L., Markmann, P., Nielsen, M.B., 1993. Re-establishment of Danish streams: restoration and maintenance measures. *Aquat. Conserv.* 3, 73–92.
- Janauer, G.A., Lanz, E., Schmidt-Mumm, U., Schmidt, B., Waidbacher, H., 2008. Aquatic macrophytes and hydro-electric power station reservoirs in regulated rivers: man-made ecological compensation structures and the "ecological potential". *Ecohydrol. Hydrobiol.* 8, 149–157.
- Johnson, P., 1988. River regulation: a regional perspective—Northumbrian water authority. *Regul. Rivers Res. Manag.* 2, 233–255.
- Johnstone, I.M., 1986. Macrophyte management: an integrated perspective. *N. Z. J. Mar. Freshw. Res.* 10, 599–614.
- Jurajda, P., 1995. Effect of channelization and regulation on fish recruitment in a flood plain river. *Regul. Rivers Res. Manag.* 10, 207–215.
- Kaenel, B.R., Uehlinger, U., 1999. Aquatic plant management: ecological effects in two streams of the Swiss plateau. *Hydrobiologia* 415, 257–263.
- Kenny, A.J., Rees, H.L., 1996. The effects of marine gravel extraction on the macrobenthos: results 2 years post-dredging. *Mar. Pollut. Bull.* 32, 615–622.
- Kruk, A., 2007. Role of habitat degradation in determining fish distribution and abundance along the lowland Warta River, Poland. *J. Appl. Ichthyol.* 23, 9–18.
- Leitão, S., Moreira-Santos, M., Van den Brink, P.J., Ribeiro, R., Cerejeira, J.M., Sousa, J.P., 2014. Ethoprophos fate on soil-water interface and effects on non-target terrestrial and aquatic biota under Mediterranean crop-based scenarios. *Ecotoxicol. Environ. Saf.* 103, 36–44.
- Lubke, R.A., Reavell, P.E., Dye, P.J., 1984. The effects of dredging on the macrophytic vegetation of the Boro river, Okavango delta, Botswana. *Biol. Conserv.* 30, 211–236.
- Maćkowiak, Ł., Kryszak, A., Kryszak, J., Klarzyńska, A., Cholewa, R., 2014. Roślinność skarp Kanału Grodzkiego w pierwszym roku po wykonanych pracach konserwacyjnych. *Woda-Środowisko-Obszary Wiejskie*. 14, pp. 81–101.
- Macnaughton, C.J., McLaughlin, F., Bourque, G., Senay, C., Lanthier, G., Harvey-Lavoie, S., Legendre, P., Lapointe, M., Boisclair, D., 2014. The effects of regional hydrologic alteration on fish community structure in regulated rivers. *River Res. Appl.* 33, 249–257.
- Manap, N., Voulvoulis, N., 2016. Data analysis for environmental impact of dredging. *J. Clean. Prod.* 137, 394–404.
- Mann, R.H.K., 1988. Fish and fisheries of regulated rivers in the UK. *Regul. Rivers Res. Manag.* 2, 411–424.
- Marcinkowski, P., Grygoruk, M., 2017. Long-term downstream effects of a dam on a lowland river flow regime: case study of the upper Narew. *Water* 9, 783.
- McCabe, G.T., Hinton Jr., S.A., Emmett, R.L., 1988. Benthic invertebrates and sediment characteristic in a shallow navigation channel of the Lower Columbia River, before and after dredging. *Northwest Science* 72, 116–126.
- McGurk, B., 1989. Predicting stream temperature after riparian vegetation removal. General Technical Report 1989, PSW-110. US Department of Agriculture, Forest Service.
- Monahan, C., Caffrey, J.M., 1996. The effect of weed control practices on macroinvertebrate communities in Irish canals. *Hydrobiologia* 340, 205–211.
- Morgan, N.C., Maitland, P.S., 1997. Conservation Management of Freshwater Habitats: Lakes, Rivers and Wetlands. Kluwer Academic Publishing, Boston.
- Moyle, P.B., 1995. Conservation of native freshwater fishes in the Mediterranean-type climate of California, USA: a review. *Biol. Conserv.* 72, 271–279.
- Munn, M.D., Brusven, M.A., 1991. Benthic invertebrate communities in nonregulated and regulated waters of the Clearwater River, Idaho, USA. *River Res. Appl.* 6, 1–11.
- Murat-Błażejewska, S., Sojka, M., 2011. Ocena struktury ichtiofauny oraz stanu hydromorfologicznego i fizykochemicznego rzeki w aspekcie dobrego stanu ekologicznego. *Nauka Przyroda Technologia* 5, 90.
- Murphy, K.J., Fox, A.M., Hanbury, R.G., 1987. A multivariate assessment of plant management impacts on macrophyte communities in a Scottish Canal. *J. Appl. Ecol.* 24, 1063–1079.
- Navarro-Llaser, Caliz, Baeza, D., de las Heras, J., 2010. Assessment of regulated rivers with indices based on macroinvertebrates, fish and riparian forest in the southeast of Spain. *Ecol. Indic.* 10, 935–942.
- de Nie, H.W., 1984. The decrease in aquatic vegetation in Europe and its consequences for fish populations. EIFAC, Occasional Paper No. 19.
- Nilsson, C., Jansson, R., 1995. Floristic differences between riparian corridors of regulated and free-flowing boreal rivers. *Regul. Rivers Res. Manag.* 11, 55–66.
- N'Zi, K.G., Yao, S.S., Bi, G.G., Ndouba, V., 2015. Update of ichthyofauna diversity and ecological status of a coastal river Nero (cote d'Ivoire - West Africa). *Saudi J. Biomed. Sci.* 22, 265–273.
- Obande, R.A., Dambo, A., Adah, P.M., 2014. Migration in fishes: a review. *Nigerian J. Fish. Aquac.* 2, 1–5.
- Ohimain, E.I., Imoibe, T.O.T., Benka-Coker, M.O., 2005. The impact of dredging on macrobenthic invertebrates in a tributary of the Warri River, Niger delta. *Afr. J. Aquat. Sci.* 30, 49–53.
- Old, G.H., Naden, P.S., Ramenshwaran, P., Acreman, M.C., Baker, S., Edwards, F.K., Sorensen, J.P.R., Mountford, O., Goody, D.C., Stratford, C.J., Scarlett, P.M., Newman, J.R., Neal, M., 2014. Instream and riparian implications of weed cutting in a chalk river. *Ecol. Eng.* 71, 290–300.
- Ożgo, M., Urbańska, M., Biereżnoj-Bazille, U., 2016. Risk associated with river maintenance outside protected areas – the case of *Unio crassus* in a tributary of Biebrza river, NE Poland. *Folia Malacologica* 24, 41–42.

- Pan, B.Z., Wang, Z.Y., Li, Z.W., Lu, Y.J., Yang, W.J., Li, Y.P., 2015. Macroinvertebrate assemblages in relation to environments in the West River, with implications for management of rivers affected by channel regulation projects. *Quat. Int.* 384, 180–185.
- Paukert, C., Schloesser, J., Fischer, J., Eitzmann, J., Pitts, K., Thornbrugh, D., 2008. Effect of instream sand dredging on fish communities in the Kansas River, USA: current and historical perspectives. *J. Freshw. Ecol.* 623–633.
- Pedersen, M.L., Baatrup-Pedersen, A., Rorth, F.R., Madsen, T.V., Larsen, S.E., 2011. Short-term impacts of weed cutting on physical habitats in lowland rivers - the importance of initial environmental conditions. *Pol. J. Environ. Stud.* 20, 1271–1280.
- Pepper, A.T., Rickard, C.E., 2009. Works in the river channel. In: Ackers, J.C., Rickard, C.J., Gill, D.S. (Eds.), *Fluvial Design Guide*. Environment Agency, Bristol, pp. 1–36.
- Petts, G.E., 1988. Regulated rivers in the United Kingdom. *Regul. Rivers Res. Manag.* 2, 201–220.
- Petts, G., Armitage, P., Castella, E., 1993. Physical habitat changes and macroinvertebrate response to river regulation: the river Rede, UK. *Regul. Rivers Res. Manag.* 8, 167–178.
- Plaska, W., Kurzątkowska, A., Stępień, E., Buczyńska, E., Pakulnicka, J., Szlauer-Lukaszewska, A., Zawal, A., 2016. The effect of dredging of a small lowland river on aquatic Heteroptera. *Ann. Zool. Fenn.* 53, 139–153.
- Poff, N.L., Zimmerman, J.K.H., 2010. Ecological responses to altered flow regimes: a literature review to inform the science and management of environmental flows. *Freshw. Biol.* 55, 194–205.
- Pool, T.K., Olden, J.D., 2015. Assessing long-term fish responses and short-term solutions to flow regulation in a dryland river basin. *Ecol. Freshw. Fish* 24, 56–66.
- Principe, R.E., del Corigliano, M.C., 2006. Benthic, drifting and marginal macroinvertebrate assemblages in a lowland river: temporal and spatial variations and size structure. *Hydrobiologia* 553, 303–317.
- Prus, P., Popek, Z., Pawlaczyk, P., 2017. Dobry praktyki utrzymania rzek. WWF Polska. (URL: http://www.kp.org.pl/pdf/2017-08-01_dobre_praktyki_utrzymania_rzek.pdf; accessed on 02.11.2017).
- Punchard, N.T., Perrow, M.R., Jowitt, A.J.D., 2008. Fish habitat associations, community structure, density and biomass in natural and channelised lowland streams in the catchment of the River Wensum, UK. In: Cowx, I.G. (Ed.), *Management and Ecology of River Fisheries*. Wiley-Blackwell, Chichester, pp. 143–157.
- Rabeni, C.F., Jacobson, R.B., 1993. The importance of fluvial hydraulics to fish-habitat restoration in low-gradient alluvial streams. *Freshw. Biol.* 29, 211–220.
- Rader, R.B., Ward, J.V., 1988. Influence of regulation on environmental conditions and the macroinvertebrate community in the Upper Colorado River. *Regul. Rivers Res. Manag.* 2, 597–618.
- Rader, R.B., Voelz, N.J., Ward, J.V., 2008. Post-flood recovery of a macroinvertebrate community in a regulated river: resilience of an anthropogenically altered ecosystem. *Restor. Ecol.* 16, 24–33.
- Rechulicz, J., Gorzel, M., Kornijów, R., 2004. Impact of channelization of a small lowland river on biodiversity of fish communities. *Teka Komisji Ochrony Kształtowania Środowiska Przyrodniczego* 1, 184–189.
- Scheffer, M., Bascompte, J., Brock, W.A., Brovkin, V., Carpenter, S.R., Dakos, V., Held, H., van Nes, E.H., Rietkerk, M., Sugihara, G., 2009. Early-warning signals for critical transitions. *Nature* 461, 53–59.
- Schoelynck, J., Creëlle, S., Buis, K., De Mulder, R., Emsens, W.J., Hein, T., Meire, D., Meire, P., Okruszko, T., Preiner, S., Roldan Gonzalez, R., Silinski, A., Temmerman, S., Troch, P., Van Oyen, T., Verschoren, V., Visser, F., Wang, C., Wolters, J.W., Folkard, A., 2017. What is a macrophyte patch? Patch identification in aquatic ecosystems and guidelines for consistent delineation. *Ecohydrol. Hydrobiol.* <https://doi.org/10.1016/j.ecohyd.2017.10.005>.
- Seppelt, R., Dormann, C.F., Eppink, F.V., Lautenbach, S., Schmidt, S., 2011. A quantitative review of ecosystem service studies: approaches, shortcomings and the road ahead. *J. Appl. Ecol.* 48, 630–636.
- Sierka, E., Tomczak, A., 2013. Macrophytes in the assessment of river ecological condition on the example of Pszczynka River (Silesian Upland). *Environmental and Socio-economic Studies* 1 (3), 14–20.
- Smokorowski, K.E., Pratt, T.C., 2007. Effect of a change in physical structure and cover on fish and fish habitat in freshwater ecosystems – a review and meta-analysis. *Environ. Rev.* 15, 15–41.
- Suding, K.N., Hobbs, R.J., 2009. Threshold models in restoration and conservation: a developing framework. *Trends Ecol. Evol.* 24, 271–279.
- Swales, S., 1982a. Impacts of weed-cutting on fisheries: an experimental study in a small lowland river. *Aquac. Res.* 13, 125–137.
- Swales, S., 1982b. A 'Before and After' study of the effects of land drainage works on fish stocks in the upper reaches of a lowland river. *Aquac. Res.* 13, 105–114.
- Swales, S., 1988. Fish populations of a small lowland channelized river in England subject to long-term river maintenance and management works. *Regul. Rivers Res. Manag.* 2, 493–506.
- Szlauer-Lukaszewska, A., 2015. Effects of river bed regulation on habitats and communities: a case study for ostracods in the Oder River, Poland. *Int. Rev. Hydrobiol.* 100, 69–78.
- Szozkiewicz, K., Jusik, S., Ławniczak, A.E., Zgoła, T., 2010. Macrophyte development in unimpacted lowland rivers in Poland. *Hydrobiologia* 656, 117–131.
- Taylor, C.M., Millican, D.S., Roberts, M.E., Stack, W.T., 2008. Long-term change to fish assemblages and the flow regime in a southeastern U.S. river system after extensive aquatic ecosystem fragmentation. *Ecography* 31, 787–797.
- Teixeira-de Mello, F., Vidal, N., Eguren, G., Loureiro, M., 2015. Length-weight relationships of 21 fish species from the lower section of the Santa Lucía River basin (Canelones-Montevideo, Uruguay). *J. Appl. Ichthyol.* 24, 491–492.
- Trepel, M., Holsten, B., Kieckbusch, J., Otten, I., Pieper, F., 2003. Influence of macrophytes on water level and flood dynamics in a riverine wetland in Northern Germany. International conference 'Towards natural flood reduction strategies'. Warsaw University of Life Sciences – SGGW, Warsaw 6–13 September 2003.
- Vereecken, H., Baetens, J., Viaene, P., Mostaert, F., Meire, P., 2006. Ecological management of aquatic plants: effects in lowland streams. *Hydrobiologia* 570, 205–210.
- Voelz, N.J., Ward, J.V., 1990. Macroinvertebrate responses along a complex regulated stream environmental gradient. *Regul. Rivers Res. Manag.* 5, 365–374.
- Walker, K.F., 1985. A review of the ecological effects of river regulation in Australia. *Hydrobiologia* 125, 111–129.
- Watson, A.M., Ormerod, S.J., 2004. The distribution of three uncommon freshwater gastropods in the drainage of British grazing marshes. *Biol. Conserv.* 118, 455–466.
- Wenger, A.S., Harvey, E., Wilson, S., Rawson, C., Newman, S.J., Clarke, D., Saunders, B.J., Browne, N., Travers, M.J., McIlwain, J.L., Erftmeijer, P.L.A., Hobbs, J.P.A., Mclean, D., Depczynski, M., Evans, R.D., 2017. A critical analysis of the direct effects of dredging on fish. *Fish Fish.* 18, 967–985.
- Wiśniewolski, W., Gieraj, A., 2011. Regulacja rzek a ichtiofauna – skutki i środki zaradcze. In: Mizieliński, M. (Ed.), *Użytkownik rybacki 2011. Kondycja polskiego rybactwa śródlądowego*. Polski Związek Wędkarski, Warszawa, pp. 90–96.
- World Bank, 2015. Data on Gross Domestic Product. URL: <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?view=map&year=2015>.
- Zawal, A., Sulikowska-Drozd, A., Stępień, E., Jankowiak, Ł., Szlauer-Lukaszewska, A., 2016. Regeneration of the molluscan fauna of a small lowland river after dredging. *Fundam. Appl. Limnol.* 187, 281–293.